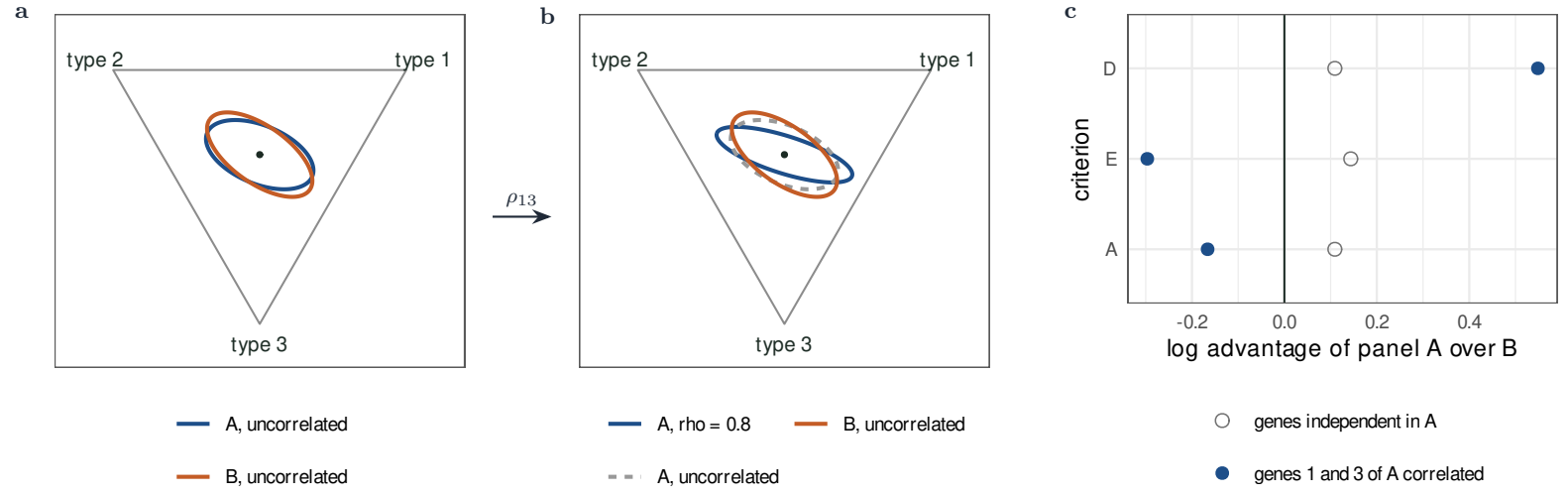
$J = 3$ types, two candidate panels of $G = 3$ genes, common within-type covariance $\boldsymbol{\Sigma}_W$: $\boldsymbol{I} = n_i \boldsymbol{K}^\top \boldsymbol{M}^\top \boldsymbol{\Sigma}_W^{-1} \boldsymbol{M}\boldsymbol{K}$,
$\mathrm{Cov}\,\hat{\boldsymbol{p}} = \boldsymbol{K}\boldsymbol{I}^{-1}\boldsymbol{K}^\top$ ($\boldsymbol{K}$ orthonormal, columns orthogonal to $\mathbf{1}$); 95 % ellipses at $\boldsymbol{p} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $n_i = 25$



$\boldsymbol{\Sigma}_W = \boldsymbol{I}$: $\boldsymbol{I} = n_i \boldsymbol{A}^\top \boldsymbol{A}$, $\boldsymbol{A} = \boldsymbol{M}\boldsymbol{K}$;
$\det \boldsymbol{I} \propto$ squared Gram volume, $\kappa(\boldsymbol{I}) = \kappa(\boldsymbol{A})^2$.
Panel A wins on D, E and A

Correlation 0.8 between
genes 1 and 3 of panel A:
smaller area (D prefers A),
longer major axis (E and A prefer B)

D: $\max \det \boldsymbol{I} \leftrightarrow$ ellipse area
E: $\max \lambda_{\min}(\boldsymbol{I}) \leftrightarrow$ longest semi-axis
A: $\min \mathrm{tr}\, \boldsymbol{I}^{-1} \leftrightarrow$ sum of squared semi-axes